Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

JAVA ASSIGNMENT

NAME:Kavya Gada

SAP ID:60017230102

ROLL NO.:A018

BRANCH:AIML

BATCH:A1

DATE OF SUBMISSION:28/03/24

NAME:Kavya Gada

SAP ID: 60017230102

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60017230102
A018 - A-I
Kavya Gada

Experiment - 4

Aim / Objective :- To implement collections (Array List / vectors)

Description of Experiment :-

- Hierarchy of Collection Framework mentioning the different classes, interface

* Interfaces :-

- Collection: Represents a group of objects known as its elements. It's the root interface in the collection hierarchy
- List: All ordered collection allows duplicate elements and maintains insertion order.
- ArrayList: Resizable-array implementation of the List interface, allows fast random access.
- Linked list :- Doubly-linked list implementation of the list interface. Provides better performance for intersection.
- Vector :- Sychronized version of array list.
- Stack : Subclass of vector. Represents (LIFO) stack of objects.
- Set : A collection that contains no duplicate element.
  - Hash-set :- Implements set interface using hash table with a linked list running throughout it. Does not guarantee the order of iteration.
  - Linked Hash-set: Maintains the insertion order.
- Tree set: Implements the sorted set interface using Red-Black tree.

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- Queue :- Represents a collection designed for holding elements prior to processing. Generally orders elements in FIFO manner.
  - Priority queue : Unordered queue that uses the natural ordering of the elements or a comparator to determine the order
  - Deque : A double ended queue
    - Array deque : Resizable array implementation of the Deque interface
- Map : An object that maps keys to values cannot contain duplicate keys.
  - HashMap : Implements map interface using hash-table
  - Linked Hash-Map : Implements maps interface using hash-table and linked list.
  - Tree map : Implements the sorted map interface using Red-Black tree
  - Hash table Synchronized version of HashMap

- Benefits of collections :-

1. Organization and management :- They provide a structured way to organize and manage ~~of~~ group of objects.
2. Flexibility : It offers a wide range of data structures and interfaces.
3. Code reusability :- By providing standard interface and implementations, it promotes code reuse
4. Performance :- They are optimized for performance with various data structures design to offer efficient

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Operations.

5. Type Safety :- It allows developers to specify the type of object that a collection can contain.

6. Thread Safety :- It allows concurrent access to collections from multiple threads.

Conclusion :- Java collections offers developers structured organization, flexibility and performance optimizations, enabling efficient data management and manipulations, while promoting code reusability and type safety.

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Experiment No:4

To implement collections (Array List/ Vectors)
WAP to accept students name from command line and store them in vector.

```
import java.util.*;
class VectorDemo
{
public static void main(String args[])
{
Vector v=new Vector();
        //to read Names from Command Line arguments and store
in to vector  v.add(args[0]);
v.add(args[1]);
v.add(args[2]);
v.add(args[3]);
v.add(args[4]);
Enumeration e=v.elements();
System.out.println("Elements in Vector: ");
while(e.hasMoreElements())
{
System.out.println(e.nextElement());
}

}
}
```

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```
C:\kavya\java\College Pratice Q\Exp 4 Code>javac VectorDemo.java
Note: VectorDemo.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.

C:\kavya\java\College Pratice Q\Exp 4 Code>java VectorDemo  Sudhir Nilesh Komal Surekha Hezal
Elements in Vector:
Sudhir
Nilesh
Komal
Surekha
Hezal
```

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WAP to add n strings in a vector array. Input new string and check if it is present in the vector. If present delete it else add to the vector

```
import java.util.*;
class SearchAndAddOrDeleteStringInVector
{
public static void main(String args[])
{
Vector<String> v=new Vector<String>();
        //to read Names from Command Line arguments and store
in to vector  v.add(args[0]);
v.add(args[1]);
v.add(args[2]);
v.add(args[3]);
v.add(args[4]);

System.out.println("Elements in Vector Before Operation: "+v);

        System.out.println("Enter String to be Search in Vector:");
        Scanner sc=new Scanner(System.in);
        String str=sc.next();

        if(v.contains(str))
            {
                v.removeElement(str);

}
        else
            {
                System.out.println("in else");
                v.addElement(str);
            }
```

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```
        System.out.println("Elements in Vector after Add or Removed input
String: "+v);

 }
 }
```

```
C:\kavya\java\College Pratice Q\Exp 4 Code>javac VectorDemo.java
Note: VectorDemo.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.

C:\kavya\java\College Pratice Q\Exp 4 Code>java VectorDemo  Sudhir Nilesh Komal Surekha Hezal
Elements in Vector:
Sudhir
Nilesh
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